

FRYING PAN SHOALS BUOY 41013, USA

WMO: 997030

Lat: 33.436N

Lon: 77.743W

Elev: 3

StdP: 101.29

Time Zone: -5.00 (W05)

Period: 05-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.2	3.3	-6.6	2.2	3.2	-4.7	2.5	5.0	16.7	12.7	16.0	14.0	9.7	10	0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	1.9	29.1	26.3	29.0	26.1	28.7	25.8	27.5	28.7	26.9	28.4	26.5	28.2	6.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	22.7	28.6	26.4	21.9	28.3	25.8	21.1	28.0	86.9	28.8	84.3	28.4	82.3	28.2	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.6	14.3	13.1	-0.1	29.6	2.4	0.4	-1.8	29.9	-3.2	30.1	-4.6	30.3	-6.3	30.5		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.4	12.6	13.4	15.0	18.8	22.2	25.6	26.9	27.3	25.8	22.2	18.0	16.0
	DBStd	6.04	5.13	4.27	3.82	2.45	1.88	1.55	1.00	1.01	1.42	3.13	3.74	4.46
	HDD10.0	67	35	16	7	0	0	0	0	0	0	0	2	8
	HDD18.3	624	184	143	114	23	2	0	0	0	0	9	52	97
	CDD10.0	3846	116	111	161	265	377	469	524	536	473	378	241	194
	CDD18.3	1361	6	5	10	38	121	219	266	278	223	129	41	24
	CDH23.3	10427	2	1	8	4	214	1812	2786	3074	1871	572	59	26
	CDH26.7	1650	0	0	0	0	0	209	507	730	192	12	0	0
Wind	WSAvg	6.9	7.6	7.8	7.7	6.7	6.2	6.0	6.1	5.4	6.5	7.1	7.7	7.6
Precipitation	PrecAvg	1433	104	96	104	89	104	139	162	175	157	114	87	101
	PrecMax	1769	197	265	198	152	182	282	315	333	443	276	189	201
	PrecMin	1006	34	33	28	22	48	37	84	81	41	2	9	38
	PrecStd	174	47	49	39	34	36	51	49	55	85	67	46	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.1	22.9	24.7	23.9	25.9	29.2	29.2	29.9	28.7	27.8	25.7	25.0
		MCWB	20.7	20.4	22.4	22.0	23.1	25.9	26.2	27.2	25.5	25.2	23.0	22.8
	2%	DB	22.0	21.8	22.0	23.3	25.5	29.0	29.0	29.4	28.4	26.9	24.8	24.2
		MCWB	19.9	20.1	19.5	21.0	22.7	25.9	26.1	26.6	25.1	24.3	21.9	22.0
	5%	DB	21.0	20.6	21.0	22.6	25.1	28.6	28.7	29.1	28.0	26.5	24.1	23.2
		MCWB	19.0	19.0	18.9	20.3	22.4	26.0	25.9	26.2	24.7	23.8	20.9	20.8
	10%	DB	19.9	19.5	20.2	22.0	24.6	28.1	28.4	28.9	27.7	25.9	23.1	22.1
		MCWB	17.7	17.3	18.0	19.8	21.9	25.5	25.7	26.0	24.4	22.9	19.8	19.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.5	21.1	23.4	22.8	24.4	27.6	27.5	28.7	26.8	26.0	23.6	23.5
		MCDB	23.1	22.4	24.8	23.5	25.5	28.7	28.5	29.4	28.1	27.2	25.2	24.6
	2%	WB	20.2	20.4	20.2	21.9	23.8	26.9	27.0	27.9	26.2	25.2	22.8	22.6
		MCDB	21.7	21.6	21.5	23.0	25.0	28.4	28.3	29.0	27.8	26.5	24.5	24.1
	5%	WB	19.5	19.5	19.5	21.1	23.0	26.4	26.7	27.2	25.7	24.6	21.6	21.6
		MCDB	21.0	20.9	20.8	22.2	24.4	28.2	28.2	28.7	27.5	26.2	23.5	23.1
	10%	WB	18.1	18.0	18.7	20.3	22.6	25.7	26.3	26.5	25.2	23.9	20.5	20.2
		MCDB	19.8	19.3	20.2	21.6	24.1	27.6	28.0	28.5	27.2	25.7	22.8	21.8
Mean Daily Temperature Range	5% DB	MDBR	5.6	5.6	4.5	3.0	2.2	1.7	1.7	1.9	2.0	2.9	4.2	4.8
		MCDBR	5.5	4.9	4.0	2.7	1.7	1.8	1.7	1.9	1.9	3.0	5.0	4.6
		MCWBR	5.9	5.1	4.2	3.1	2.1	2.1	2.0	2.0	2.5	2.9	4.9	5.0
	5% WB	MCDBR	5.4	4.5	3.8	3.0	1.8	1.5	1.6	1.8	2.0	3.0	5.1	4.3
		MCWBR	5.7	5.0	4.0	3.6	2.4	2.1	2.2	2.1	2.3	2.9	5.2	4.9
Clear-Sky Solar Irradiance	taub	0.341	0.349	0.372	0.401	0.451	0.476	0.507	0.488	0.430	0.390	0.358	0.346	
	taud	2.520	2.502	2.452	2.351	2.270	2.245	2.173	2.228	2.382	2.467	2.504	2.529	
	Ebn at Noon	872	903	905	888	843	818	790	800	835	846	845	845	
	Edh at Noon	85	95	108	124	136	139	149	139	114	97	85	80	
All-Sky Solar Radiation	RadAvg	2.65	3.38	4.55	5.82	6.49	6.66	6.23	5.58	4.65	3.96	2.97	2.37	
	RadStd	0.20	0.34	0.42	0.43	0.40	0.43	0.42	0.37	0.36	0.38	0.22	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+178	+120

Nomenclature: See separate page